



Chapter 4

'Animal Kingdom'

The Animal Kingdom includes a diverse group of multicellular organisms that show a wide range of body plans, habitats and adaptations.

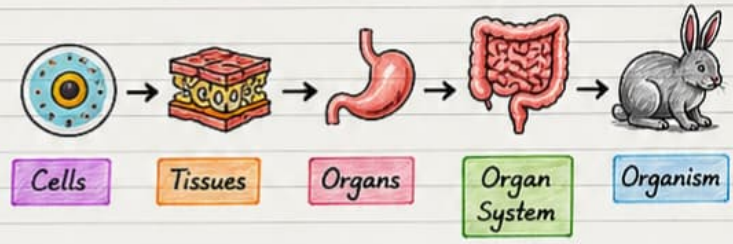
Quick Facts

- Multicellular
- Heterotrophic
- Eukaryotic
- No cell wall
- Usually motile
- Exhibit tissues/organs
- Sexual reproduction
- Well developed nervous system

1. General Characters

- Multicellular, eukaryotic organisms.
- Cells lack cell wall.
- Heterotrophic - depend on other organisms for food.
- Generally motile at some stage of life.
- Body made up of tissues, organs and organ systems.
- Sexual reproduction is common.
- Exhibit nervous coordination.
- Examples: Sponges, Hydra, Earthworm, Cockroach, Pisces, Birds, Humans etc.

Levels of Organization



2. Basis of Classification

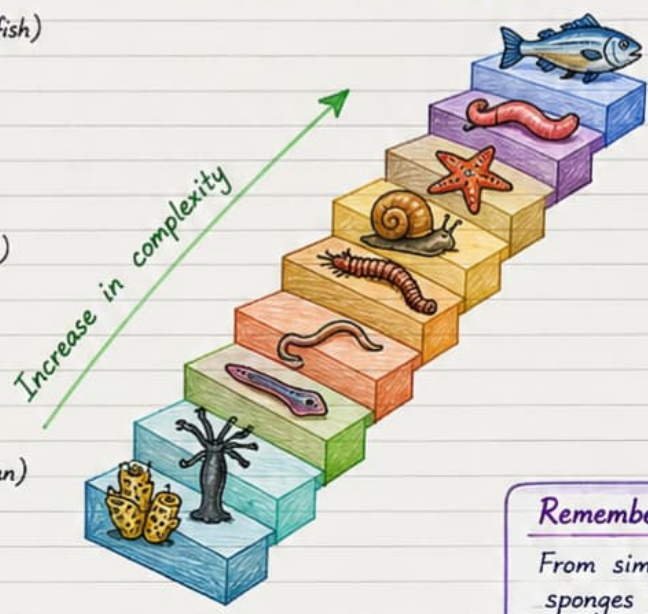
Animals are classified on the basis of -

- Level of organization
- Symmetry
- No. of germ layers
- Coelom (body cavity)
- Segmentation
- Notochord
- Habitat and other features

Simpler forms
→
Complex forms

3. Major Groups in Animal Kingdom

1. **Porifera** (Sponges)
2. **Coelenterata / Cnidaria** (Hydra, Jellyfish)
3. **Platyhelminthes** (Flatworms)
4. **Nematoda** (Roundworms)
5. **Annelida** (Segmented worms)
6. **Arthropoda** (Insects, Prawns, Spiders)
7. **Mollusca** (Snails, Octopus)
8. **Echinodermata** (Starfish, Sea urchin)
9. **Hemichordata** (Acorn worms)
10. **Chordata** (Vertebrates - Fish to Human)



4. Key Points

- Animals show great diversity in size, shape and habitat.
- They play important roles in ecosystems.
- Understanding animal diversity helps in conservation and human welfare.

Remember!

From simple sponges to complex humans - nature shows an amazing variety!



5. Key Features of Animals ✨

- Multicellular, eukaryotic organisms.
- Heterotrophic (ingest food).
- Cells lack cell wall.
- Specialized tissues, organs and organ systems.
- Usually motile at some stage.
- Well developed nervous system and sense organs.
- Exhibit sexual reproduction.

Did You Know?

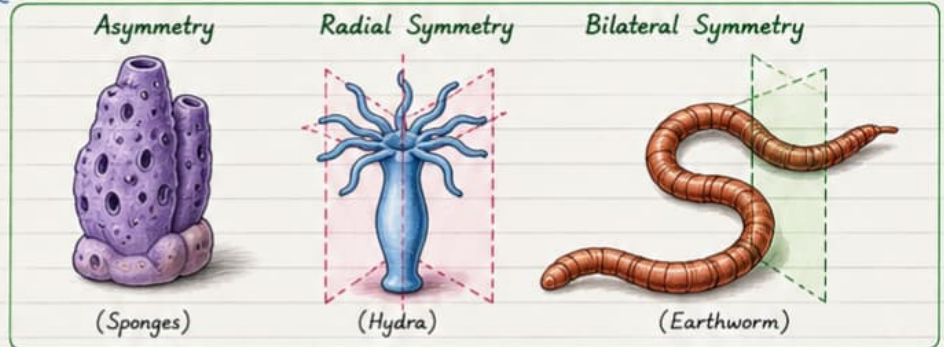
Sponges are the simplest animals without true tissues, while humans are the most complex!

6. Diploblastic vs Triploblastic

Feature	Diploblastic (2 germ layers)	Triploblastic (3 germ layers)
Germ Layers	Ectoderm, Endoderm	Ectoderm, Mesoderm, Endoderm
Level of Organization	Cellular level	Tissue/Organ level
Body Cavity	No true coelom	Coelom present (Acoelomate/Pseudocoelomate/Coelomate)
Example Phyla	Porifera, Cnidaria	Platyhelminthes, Nematoda onwards

7. Symmetry in Animals

- Asymmetry** - No plane divides into equal halves. (Example: Sponges)
- Radial Symmetry** - Many planes pass through central axis. (Example: Hydra, Starfish)
- Bilateral Symmetry** - Only one plane divides into two equal halves. (Example: Earthworm, Humans)



8. Coelom: Body Cavity

Acoelomate	Pseudocoelomate	Coelomate (True coelom)
(No body cavity) Example: Platyhelminthes	Example: Nematoda	Example: Annelida, Arthropoda, Mollusca, Echinodermata, Chordata

Quick Recall

- ★ **Acoelomate** - No cavity between body wall and gut.
- ★ **Pseudocoelomate** - Cavity present but not lined by mesoderm.
- ★ **Coelomate** - True body cavity lined by mesoderm (mesodermal origin).

9. Digestive System Types

		Example
1. Incomplete	Single opening acts as both mouth and anus.	Hydra, Platyhelminthes
2. Complete	Mouth and anus are separate.	Earthworm, Humans
3. Absent	No digestive system; food digested inside cells.	Sponges

10. Reproduction in Animals

- Mostly sexual reproduction.
- Some show asexual reproduction (e.g., Hydra, Planaria).
- Modes: Oviparous (egg laying), Viviparous (live birth), Ovoviviparous (egg hatching inside body).
- Many animals show parental care (e.g., Birds, Mammals).

NEET Tip

Remember examples with each feature - it helps in scoring fast in MCQs!

11. Examples (Phyla at a Glance)

Phylum	Common Name	Key Features	Example	Representative Diagram
1. Porifera	Sponges	Asymmetrical, no true tissues, pores for water flow.	Sycon, Spongilla	
2. Cnidaria	Coelenterates	Radial symmetry, stinging cells (cnidocytes), cavity in body.	Hydra, Jellyfish	
3. Platyhelminthes	Flatworms	Dorsoventrally flat, acoelomate, incomplete gut.	Planaria, Fasciola	
4. Nematoda	Roundworms	Unsegmented, cylindrical, pseudocoelomate.	Ascaris, Wuchereria	
5. Annelida	Segmented worms	Segmented body, true coelom, closed circulatory system.	Earthworm, Nereis	
6. Arthropoda	Insects, Prawns, Spiders	Segmented body, jointed appendages, exoskeleton.	Cockroach, Prawn	
7. Mollusca	Molluscs	Soft body, muscular foot, often calcareous shell.	Pila (snail), Octopus	
8. Echinodermata	Spiny skinned	Marine, spiny skin, radial symmetry in adults.	Starfish, Sea urchin	
9. Chordata	Animals with notochord	Notochord, dorsal hollow nerve cord, pharyngeal gill slits (in some).	Herd, Human	

Remember!

These phyla do not have a notochord at any stage of their life cycle.

Major Phyla of Non-Chordates

Key Points

- Great diversity
- Adapted to terrestrial, aquatic & parasitic habitats
- Exhibit different levels of body organization

1. Porifera (Sponges)



- Mostly marine (few in freshwater)
- Asymmetrical, no true tissues
- Body with pores (ostia) and canal system
- Spicules (calcareous/siliceous) or spongin fibers present
- Example: Sycon, Spongilla, Euspongia (bath sponge)

2. Cnidaria (Coelenterates)



- Aquatic (marine & freshwater)
- Diploblastic, radial symmetry
- Tentacles with stinging cells (cnidocytes)
- Two forms: Polyp & Medusa
- Example: Hydra, Obelia, Aurelia (jellyfish), Sea anemone

3. Platyhelminthes (Flatworms)



- Triploblastic, acoelomate, bilaterally symmetrical
- Dorsoventrally flattened body
- Incomplete digestive system
- Many are parasitic
- Example: Planaria (free living), Fasciola (liver fluke), Taenia (tapeworm)

4. Nematoda (Roundworms)



- Triploblastic, pseudocoelomate, bilaterally symmetrical
- Body cylindrical, unsegmented, tapering at both ends
- Complete digestive system
- Many are parasitic in plants & animals
- Example: Ascaris (roundworm), Wuchereria (filarial worm)

5. Annelida (Segmented Worms)



- Triploblastic, coelomate, bilaterally symmetrical
- Body divided into segments (metameres)
- Setae or parapodia may be present
- Closed circulatory system
- Example: Pheretima (earthworm), Nereis (sandworm), Hirudinaria (leech)

6. Arthropoda (Insects, Prawns, Spiders)



- Largest phylum of animal kingdom
- Triploblastic, coelomate, segmented
- Jointed appendages
- Chitinous exoskeleton
- Open circulatory system
- Example: Prawns, Crabs, Spiders, Cockroach, Butterflies

7. Mollusca (Molluscs)



- Triploblastic, coelomate, bilaterally symmetrical
- Soft, unsegmented body
- Mantle present (secretes shell in most)
- Radula (rasping organ) usually present
- Examples: Pila (snail), Unio (freshwater mussel), Octopus

8. Echinodermata (Spiny Skinned)



- Exclusively marine
- Triploblastic, coelomate, radial symmetry (adults)
- Endoskeleton of calcareous plates
- Tube feet (water vascular system) for movement
- Examples: Starfish (Asterias), Sea urchin, Sea cucumber

9. Hemichordata (Acorn Worms)



- Marine, worm-like animals
- Proboscis present
- Pharyngeal gill slits present
- Body divided into 3 parts: Proboscis, Collar & Trunk
- Example: Balanoglossus

Quick Comparison

Phylum	Level of Organization	Body Cavity	Symmetry	Habitat
Porifera	Cellular	No true coelom	Asymmetrical	Mostly marine
Cnidaria	Tissue	No true coelom	Radial	Aquatic
Platyhelminthes	Organ	Acoelomate	Bilateral	Aquatic/Parasitic
Nematoda	Organ	Pseudocoelomate	Bilateral	Aquatic/Parasitic
Annelida	Organ-system	Coelomate	Bilateral	Aquatic/terrestrial
Arthropoda	Organ-system	Coelomate	Bilateral	Terrestrial/Aquatic
Mollusca	Organ-system	Coelomate	Bilateral	Aquatic/terrestrial
Echinodermata	Organ-system	Coelomate	Radial (adult)	Marine
Hemichordata	Organ-system	Coelomate	Bilateral	Marine

NEET Tip



Learn examples with each phylum - NEET often asks example-based questions!

Remember!

All Chordates show notochord at some stage of life.

Phylum Chordata & Subphylum Vertebrata

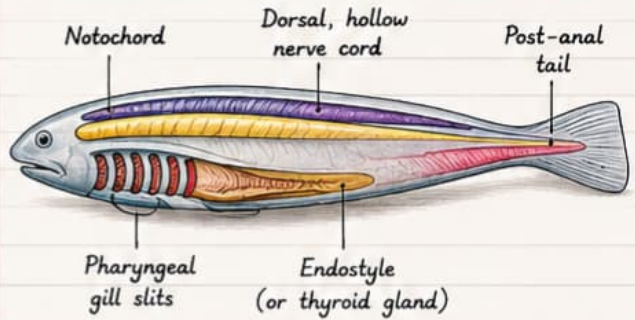
Key Points

- Notochord at some stage
- Dorsal, hollow nerve cord
- Pharyngeal gill slits
- Post-anal tail
- Endostyle or thyroid gland
- Heart ventral, closed circulatory system

1. Phylum Chordata – General Characters

- Notochord present at least during embryonic stage.
- Dorsal, hollow nerve cord.
- Pharyngeal gill slits present.
- Post-anal tail present.
- Endostyle present in pharynx.
- Heart ventral, closed circulatory system.
- Bilaterally symmetrical, triploblastic, coelomate.
- Sexes separate.

Five Key Chordate Characters (at some stage)



2. Subphylum Vertebrata – General Characters



- Notochord replaced by vertebral column.
- Well developed skull enclosing brain.
- Dorsal hollow nerve cord present.
- Paired appendages (fins or limbs) present.
- Endoskeleton of cartilage or bone.
- Advanced sense organs.
- Complex organ systems.

Did You Know?

Vertebrates are more advanced than invertebrates because of presence of vertebral column, skull and complex organs.

3. Classification of Vertebrates (Classes)

1. Pisces (Fishes)



- Aquatic
- Gills for respiration
- Fins for locomotion
- Cold-blooded

Example: Rohu, Shark

2. Amphibia (Amphibians)



- Lives both in water and on land
- Moist skin
- Larva with gills, adult with lungs & skin
- Cold-blooded

Example: Frog, Toad

3. Reptilia (Reptiles)



- Dry, scaly skin
- Lungs for respiration
- Lay shelled eggs on land
- Cold-blooded

Example: Lizard, Snake, Turtle

4. Aves (Birds)



- Body covered with feathers
- Forelimbs modified into wings
- Lungs with air sacs
- Warm-blooded
- Oviparous

Example: Pigeon, Eagle

5. Mammalia (Mammals)



- Body with hair
- Mammary glands for feeding young
- Mostly viviparous
- Warm-blooded

Example: Human, Dog, Cow

4. Comparative Features of Vertebrate Classes

Feature	Pisces	Amphibia	Reptilia	Aves	Mammalia
Habitat	Aquatic	Both aquatic & terrestrial	Terrestrial	Terrestrial (Aerial)	Terrestrial (Aquatic in some)
Body Covering	Scales	Moist skin	Dry scales	Feathers	Hair
Respiration	Gills	Gills (larva), Lungs & skin (adult)	Lungs	Lungs + Air sacs	Lungs
Heart	2 chambered	3 chambered	3 chambered (incomplete)	4 chambered (complete)	4 chambered (complete)
Temperature Regulation	Cold-blooded	Cold-blooded	Cold-blooded	Warm-blooded	Warm-blooded
Reproduction	Oviparous	Mostly Oviparous	Oviparous	Oviparous	Mostly Viviparous

5. Examples of Vertebrates

- Pisces – Rohu, Catla, Shark
- Amphibia – Frog, Toad, Salamander
- Reptilia – Lizard, Snake, Crocodile, Turtle
- Aves – Pigeon, Crow, Eagle, Ostrich
- Mammalia – Human, Cow, Dog, Dolphin, Bat

NEET Tip

Learn key characters and examples with diagrams to score better in NCERT based questions!

